Printed Pages:01 Sub Code: KOT-052

Paper Id: 232017 Roll No. | | | | | | | | | | |

B.TECH
(SEM V) THEORY EXAMINATION 2022-23
PROGRAMMING& INTERFACING WITH MICROCONTROLLERS

Time: 3 Hours *Total Marks: 100*

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

a. Define ARM.
b. What is open source platform?
c. Give few applications of Raspberry Pi.
d. What is the difference between analog circuit and digital circuits?
e. Define open framework.
f. What type of actuator is used in Arduino?
g. Define SQLite.
h. What is XML stand for?
i. What is difference TCP and UDP?
j. What is the future scope of IoT?

SECTION B

2. Attempt any *three* of the following: **10 x 3 = 30**

a. What is process code? Explain variables and flow control.
b. Write the steps for problem solving in programming? Explain DAC.
c. Explain the concept of serial peripheral interface (SPI). Give few example of SPI.
d. Why is interfacing needed? Explain I/O interfacing in detail.
e. Explain building peer to peer communication system using Bluetooth &WiFi.

SECTION C

3. Attempt any *one* part of the following: **10 x 1 = 10**

a. Explain sensor characterization. How can we characterize sensor performance?
b. What are the ARM devices? Discuss real world interfacing.

4. Attempt any *one* part of the following: **10 x 1 = 10**

a. Discuss hardware framework. What is a Raspberry Pi used for?
b. Explain Raspberry Pi Technology.

5. Attempt any *one* part of the following: **10 x 1 = 10**

a. How do hardware components communicate with each other?
b. Explain XML Interface with suitable example.

6. Attempt any *one* part of the following: **10 x 1 = 10**

a. Explain PHP. What is difference between SQL and SQLite?
b. Explain Live Network feeds (push and pull).

7. Attempt any *one* part of the following: **10 x 1 = 10**

a. Explain UDP in detail.
b. Write short notes on (i)Internet of things (ii) TCP/IP model

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